1 Introduction to agents and their world

This chapter covers

- Defining the concept of agents
- Differentiating the components of an agent
- Analyzing the rise of the agent era: Why agents?
- Peeling back the AI interface
- Navigating the agent landscape

The agent isn't a new concept in machine learning and artificial intelligence (AI). In reinforcement learning, for instance, the word *agent* denotes an active decision-making and learning intelligence. In other areas, the word *agent* aligns more with an automated application or software that does something on your behalf.

1.1 Defining agents

You can consult any online dictionary to find the definition of an agent. The *Merriam-Webster Dictionary* defines it this way (www.merriam-webster.com/dictionary/agent):

- One that acts or exerts power
- Something that produces or can produce an effect
- A means or instrument by which a guiding intelligence achieves a result

The word *agent* in our journey to build powerful agents in this book uses this dictionary definition. That also means the term *assistant* will be synonymous with *agent.* Tools like OpenAI's GPT Assistants will also fall under the AI agent blanket. OpenAI avoids the word *agent* because of the